

AWS DevOps Engineer Intern Assignment Report

Student Information

Name: Laxman Adhikari

Email: adhikarilaxman441@gmail.com

Assignment: AWS DevOps Engineer Intern Assignment

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1. Objective

The objective of this assignment was to deploy a simple static website on an AWS EC2 Ubuntu instance using the Nginx web server. The assignment also involved learning basic Linux administration, configuring AWS networking, using Git and GitHub for version control, and documenting the complete deployment process.

2. AWS Services Used

Amazon EC2

Amazon EC2 (Elastic Compute Cloud) was used to create a virtual Ubuntu Linux server for hosting the website.

Configuration

- Operating System: Ubuntu 22.04 LTS
 - Instance Type: t2.micro (AWS Free Tier)
 - Storage: 8 GB
 - Network: Default VPC
-

Security Group

A Security Group was created with the following rules:

Type	Port	Purpose
SSH	22	Remote Login
HTTP	80	Website Access

3. EC2 Instance Setup

The following steps were performed:

1. Logged into the AWS Management Console.
2. Created a new Ubuntu EC2 instance.
3. Generated and downloaded a key pair (.pem file).
4. Configured the Security Group.
5. Launched the instance.
6. Connected to the server using SSH.

SSH Command:

```
ssh -i assignment.pem ubuntu@<EC2-Public-IP>
```

The EC2 instance was successfully launched and accessed through SSH.

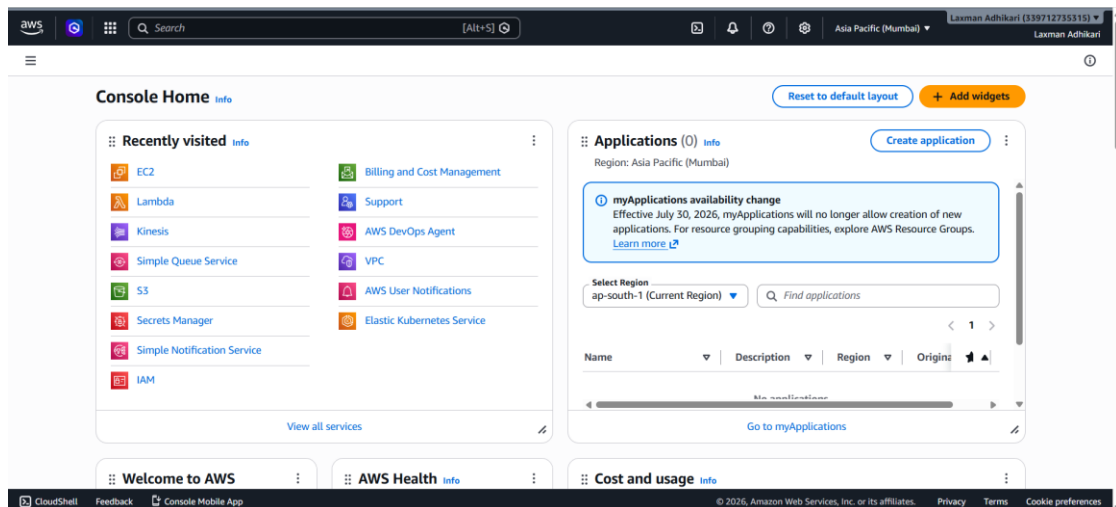


Figure 01: AWS Management Console - EC2 Dashboard

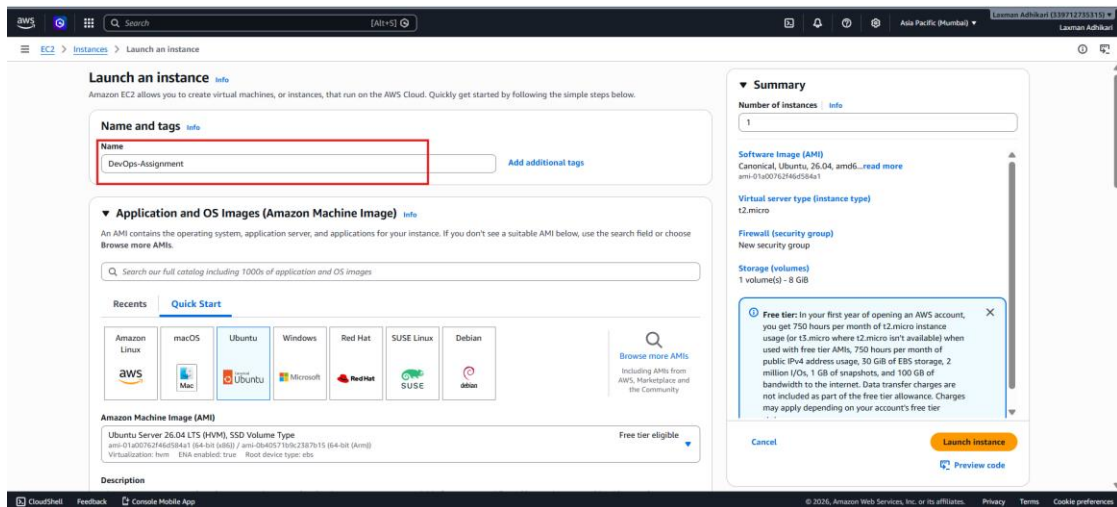


Figure 02: Launching a New EC2 Instance

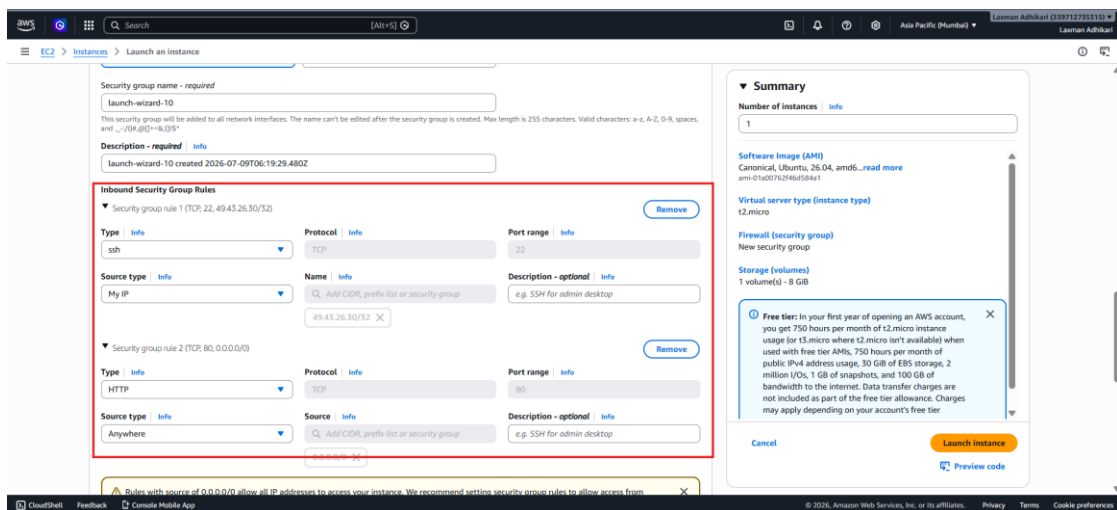


Figure 03: EC2 Instance Configuration

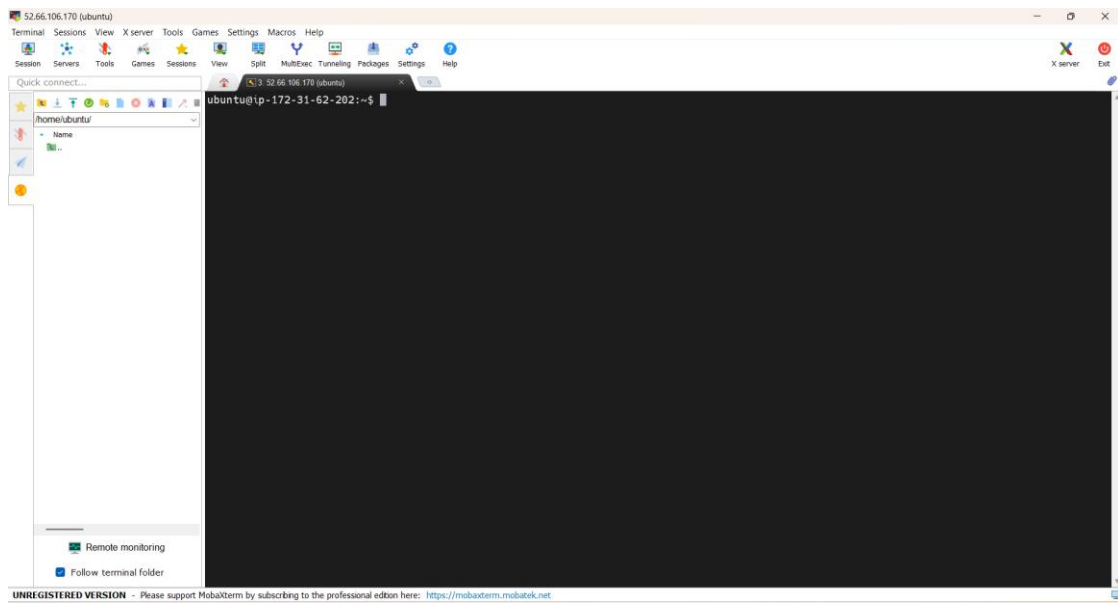


Figure 04: Security Group Configuration (SSH & HTTP rules)

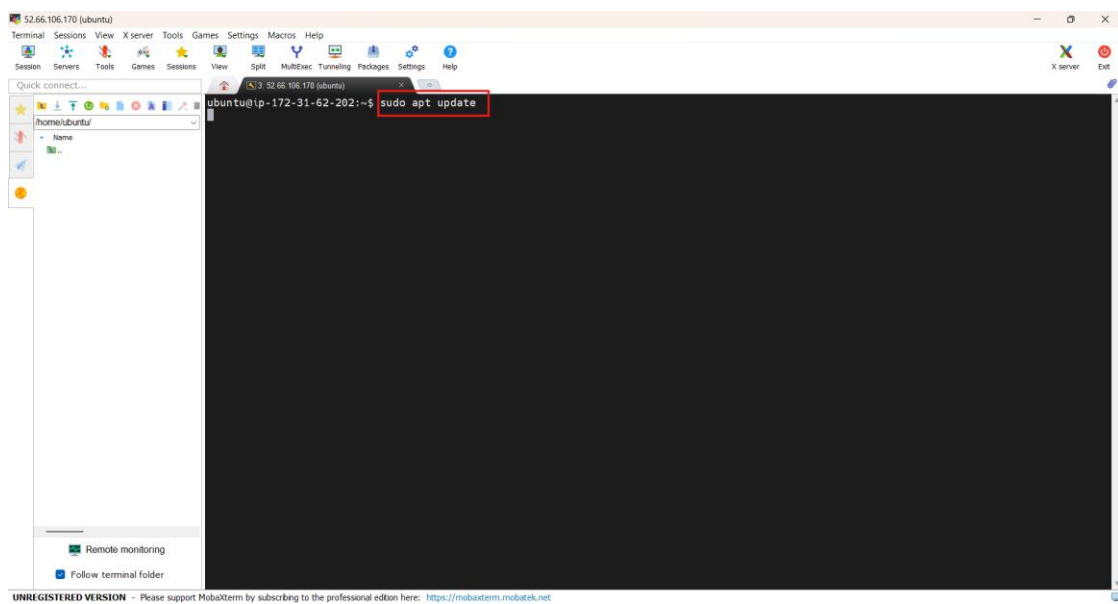


Figure 05: EC2 Instance Running with Key Pair

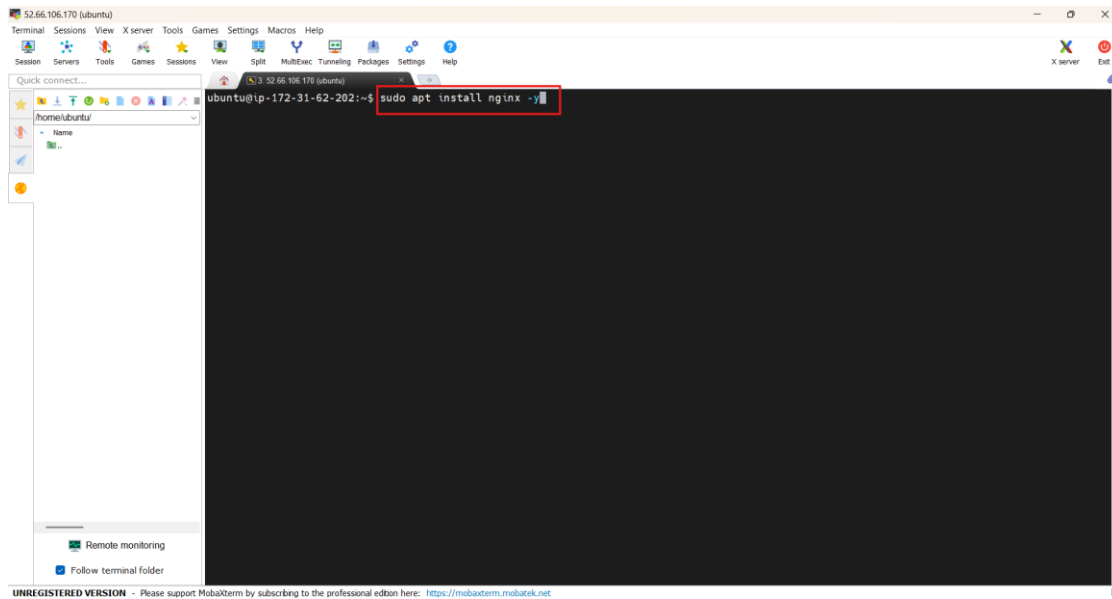


Figure 06: SSH Login to EC2 Instance

4. Linux Commands Used

The following Linux commands were executed during the assignment.

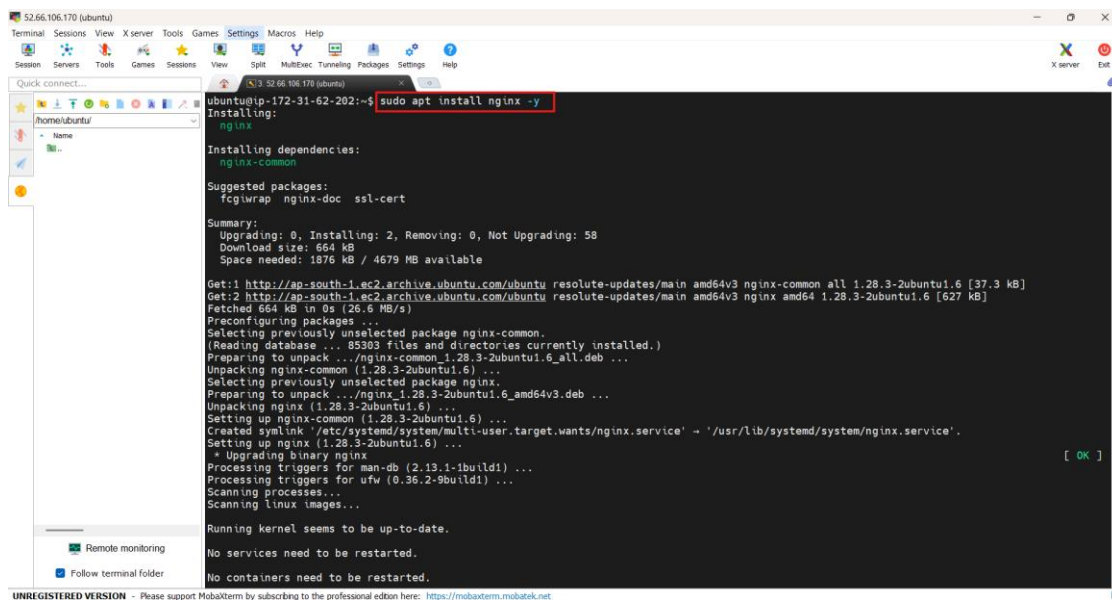
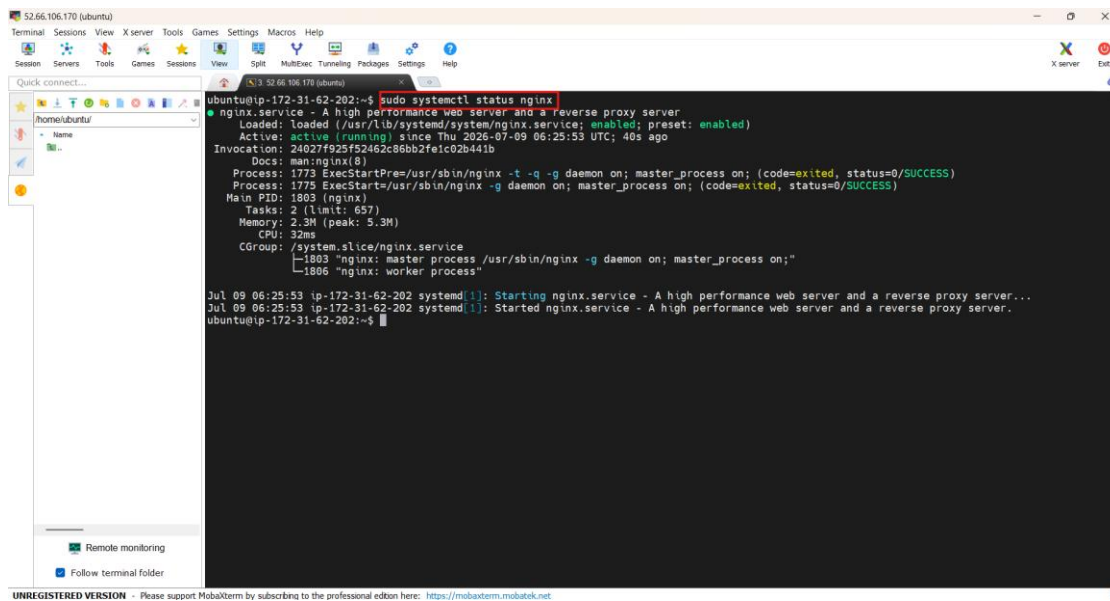


Figure 07: Output of sudo apt update

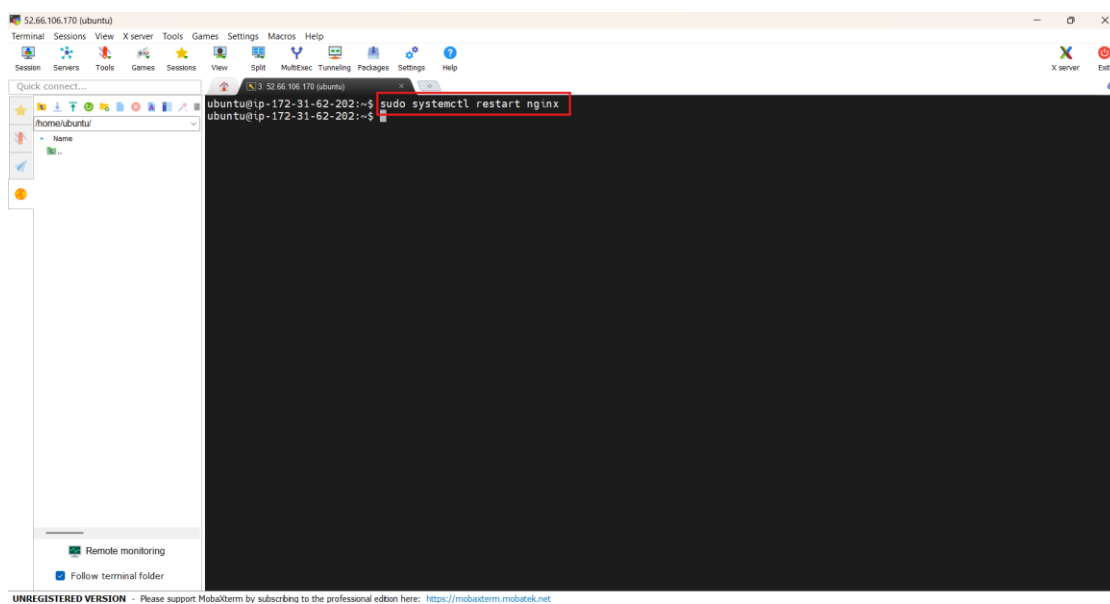


The screenshot shows a terminal window with the command `sudo systemctl status nginx` executed. The output displays the status of the nginx service, including its loaded state, active status, and process details. The terminal window has a menu bar with options like Terminal, Sessions, View, X server, Tools, Games, Settings, Macros, and Help. A sidebar on the left shows a 'Quick connect...' section with a 'home/ubuntu' entry. At the bottom, there is a 'Remote monitoring' section with a 'Follow terminal folder' option. A watermark at the bottom left reads 'UNREGISTERED VERSION - Please support MobaXterm by subscribing to the professional edition here: https://mobaxterm.mobatek.net'.

```
ubuntu@ip-172-31-62-202:~$ sudo systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Thu 2026-07-09 06:25:53 UTC; 40s ago
     Invocation: 24027f925f52462c86bb2fe1c02b441b
       Docs: man:nginx(8)
    Process: 1773 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
    Process: 1775 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
   Main PID: 1803 (nginx)
      Tasks: 2 (limit: 657)
     Memory: 2.3M (peak: 5.3M)
        CPU: 32ms
   CGroup: /system.slice/nginx.service
           └─1803 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
              └─1806 "nginx: worker process"

Jul 09 06:25:53 ip-172-31-62-202 systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
Jul 09 06:25:53 ip-172-31-62-202 systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.
ubuntu@ip-172-31-62-202:~$
```

Figure 08: Output of `sudo apt upgrade -y`



The screenshot shows a terminal window with the command `sudo systemctl restart nginx` entered. The terminal window has the same menu bar and sidebar as Figure 08. A watermark at the bottom left reads 'UNREGISTERED VERSION - Please support MobaXterm by subscribing to the professional edition here: https://mobaxterm.mobatek.net'.

```
ubuntu@ip-172-31-62-202:~$ sudo systemctl restart nginx
ubuntu@ip-172-31-62-202:~$
```

Figure 09: Installing Nginx (`sudo apt install nginx -y`)

```
ubuntu@ip-172-31-62-202:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
/dev/root        6.7G  2.3G  4.4G  34% /
dev/shm          476M   0  476M   0% /dev/shm
tmpfs            191M  908K  190M   1% /run
tmpfs            476M   0  476M   0% /tmp
none            1.0M   0   1.0M   0% /run/credentials/systemd-journald.service
none            1.0M   0   1.0M   0% /run/credentials/systemd-resolved.service
/dev/xvda13      989M  96M  827M  11% /boot
/dev/xvda15      105M  6.3M   99M   7% /boot/efi
none            1.0M   0   1.0M   0% /run/credentials/systemd-networkd.service
none            1.0M   0   1.0M   0% /run/credentials/getty@tty1.service
none            1.0M   0   1.0M   0% /run/credentials/serial-getty@ttyS0.service
tmpfs            96M   8.0K   96M   1% /run/user/1000
```

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Figure 10: Checking Nginx Status (`sudo systemctl status nginx`)

```
ubuntu@ip-172-31-62-202:~$ free -h
total        used        free      shared  buff/cache   available
Mem:          951Mi       331Mi       168Mi       1.8Mi       570Mi       620Mi
Swap:           0B           0B           0B
```

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Figure 11: Restarting Nginx (`sudo systemctl restart nginx`)



Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

Figure 12: Disk Usage Output (df -h)

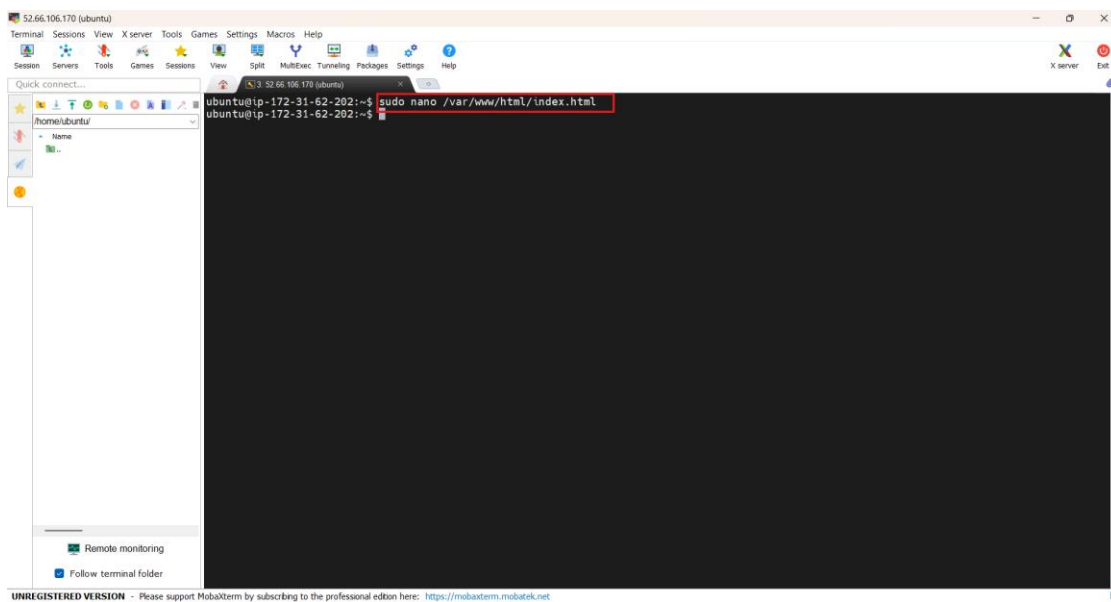


Figure 13: Memory Usage Output (free -h)

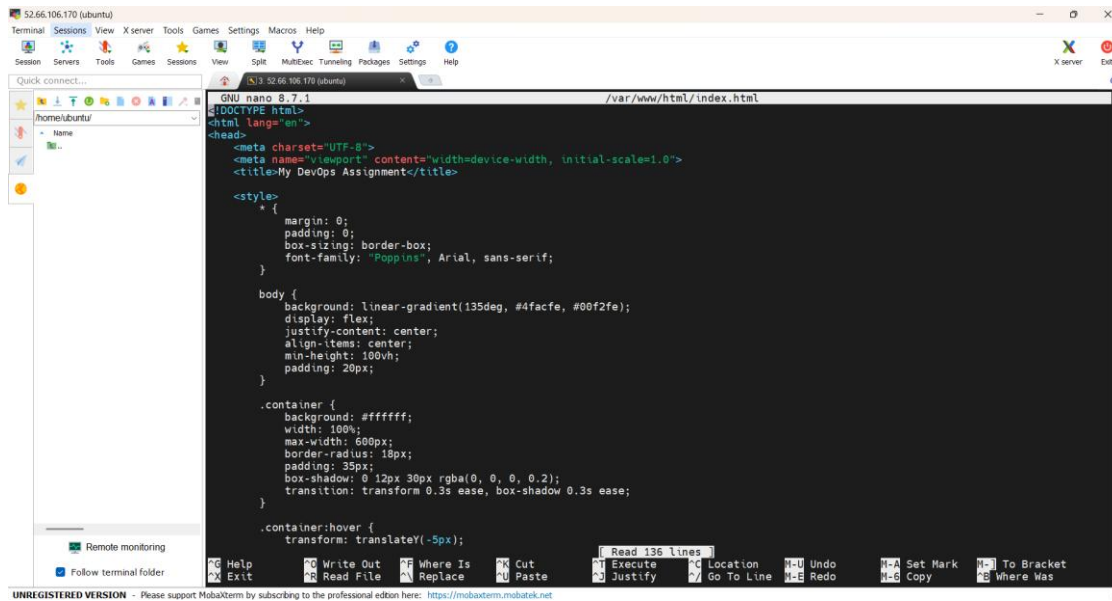


Figure 14: Running Processes Output (ps aux)

Update Package Repository

```
sudo apt update
```

```
sudo apt upgrade -y
```

Install Nginx

```
sudo apt install nginx -y
```

Check Nginx Status

```
sudo systemctl status nginx
```

Restart Nginx

```
sudo systemctl restart nginx
```

Check Disk Usage

```
df -h
```

Check Memory Usage

```
free -h
```

Display Running Processes

```
ps aux
```

5. Website Deployment

A simple HTML webpage was created containing:

- Name
- College
- Branch
- Email
- Current Date

The default Nginx webpage was replaced by editing the following file:

```
sudo nano /var/www/html/index.html
```

The website was successfully deployed and accessed using:

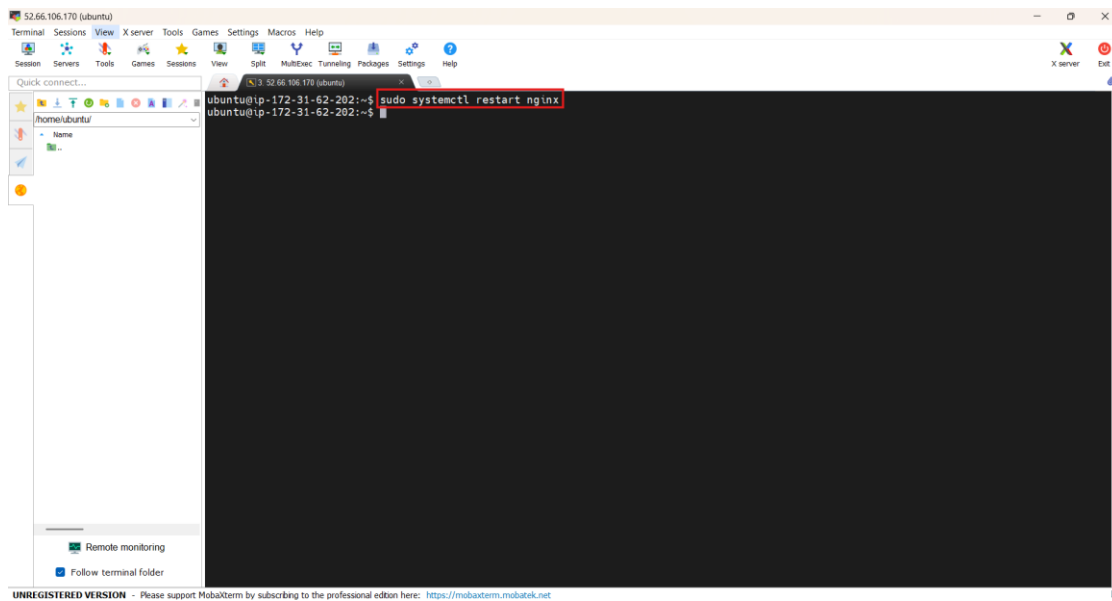


Figure 15: Editing index.html using nano

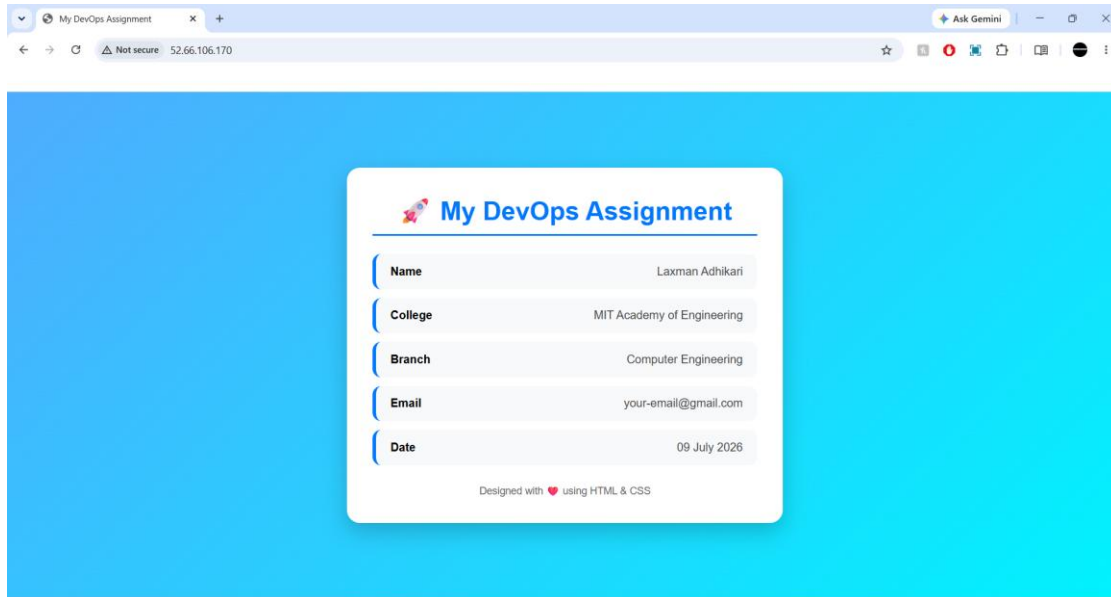


Figure 16: Website Running Successfully in Browser

`http://<EC2-Public-IP>`

6. Git and GitHub

A GitHub repository was created to store the project files.

The repository contains:

- index.html
- README.md

Git commands used:

```
git init
git add .
git commit -m "Initial Commit"
git branch -M main
git remote add origin <Repository URL>
git push -u origin main
```

GitHub Repository:

`https://github.com/adhikarilaxman/aws-devops-assignment`

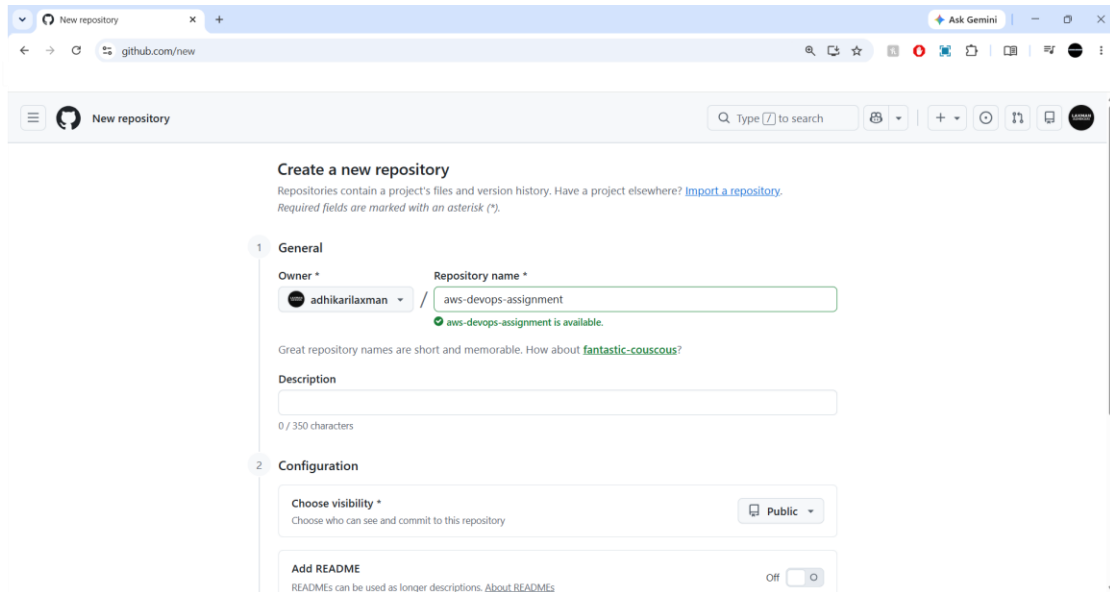


Figure 17: Initializing Git Repository (git init)

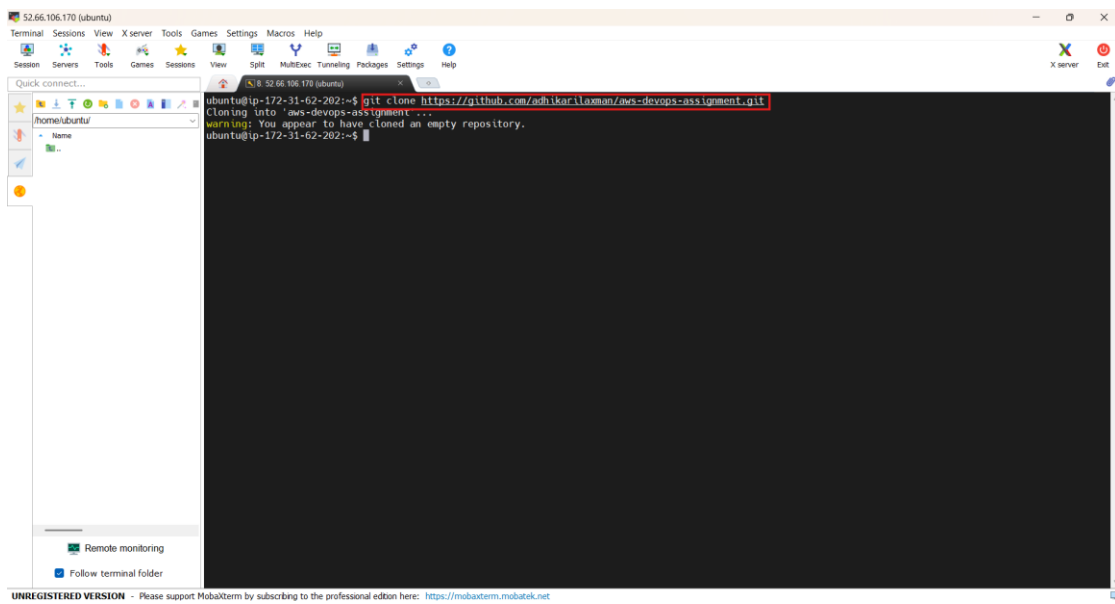


Figure 18: Adding and Committing Files (git add & git commit)

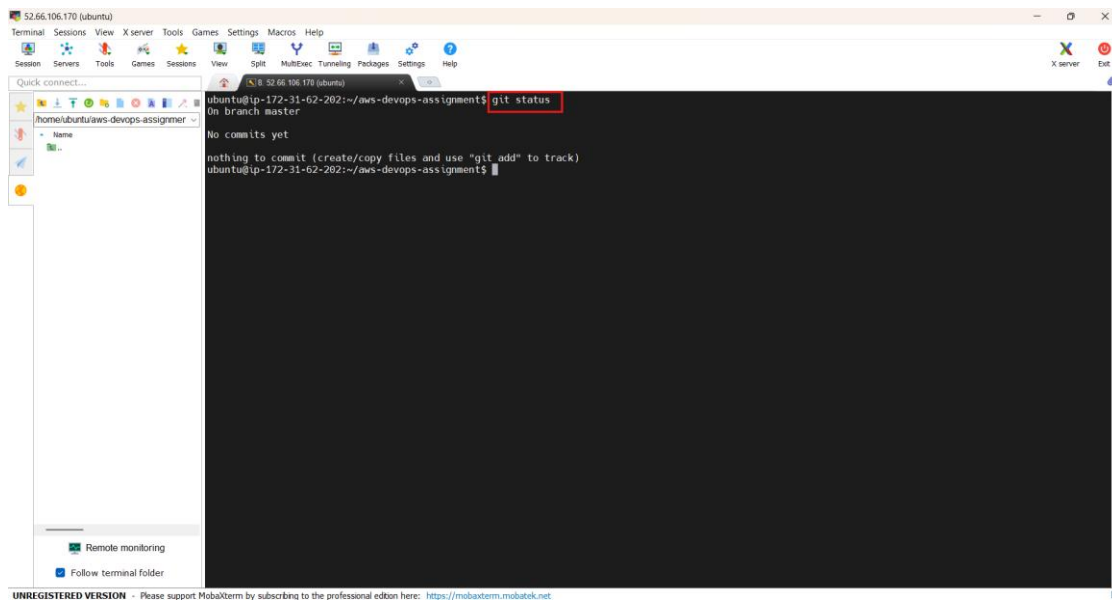


Figure 19: Pushing Code to GitHub (git push)

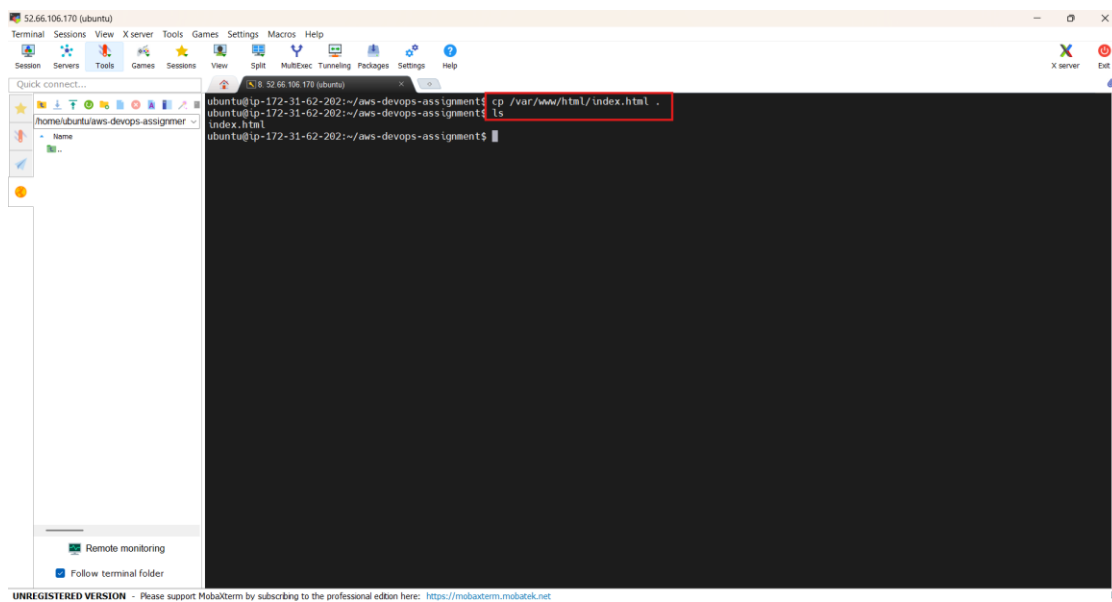


Figure 20: GitHub Repository Page

7. Problems Faced

Problem 1

Unable to access the website.

Reason

HTTP port (80) was not allowed in the Security Group.

Solution

Added an inbound rule allowing HTTP traffic on port 80.

Problem 2

Git clone failed because the folder already existed.

Reason

The repository directory already existed on the EC2 instance.

Solution

Entered the existing directory instead of cloning again, or removed the directory and cloned the repository again.

Problem 3

Permission denied while editing website files.

Solution

Used the sudo command to modify files inside /var/www/html.

8. Learnings

During this assignment, I learned:

- How to launch an AWS EC2 instance.
 - How to connect to an EC2 instance using SSH.
 - Basic Linux administration commands.
 - Installation and management of the Nginx web server.
 - Hosting a static website on Linux.
 - Configuring AWS Security Groups.
 - Using Git and GitHub for version control.
 - Documenting cloud deployment activities.
-

9. Time Taken

Activity	Time
EC2 Setup	5 Minutes
Linux Configuration	10 Minutes
Website Deployment	10 Minutes
GitHub Repository	10 Minutes
Documentation	25 Minutes

Total Time: Approximately 1 Hours

10. Bonus Task

Docker was installed on the EC2 instance and the official hello-world container was executed successfully.

Commands used:

```
sudo apt install docker.io -y
sudo systemctl start docker
sudo systemctl enable docker
sudo docker run hello-world
```

This verified that Docker was installed and functioning correctly.

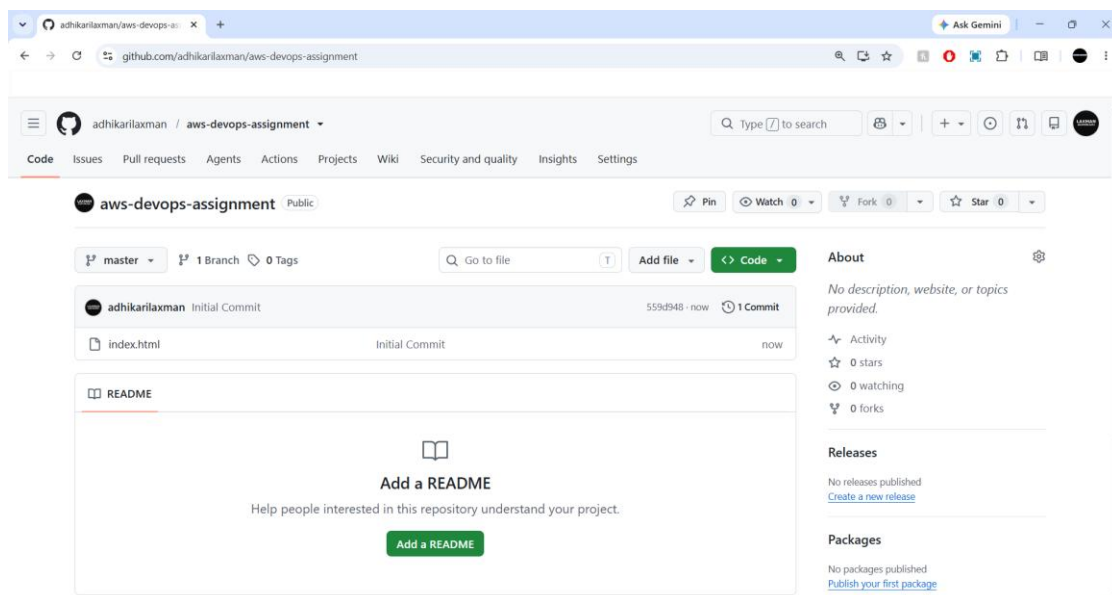
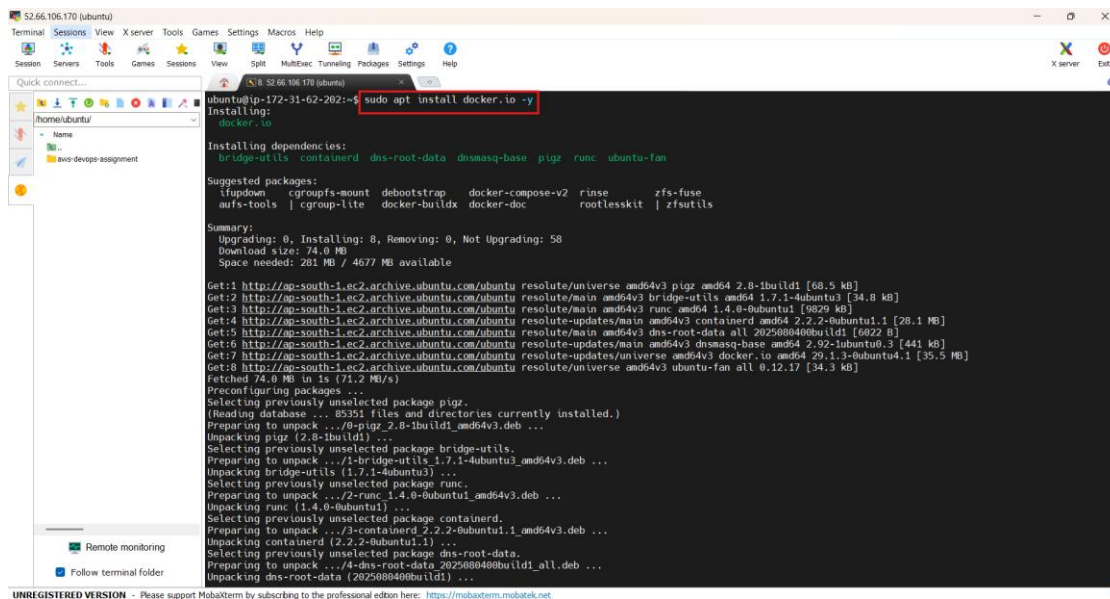


Figure 21: Installing Docker on EC2 Instance



```
52.66.106.170 (ubuntu)
Terminal Sessions View X server Tools Games Settings Macros Help
Session Servers Tools Games Sessions View Split Multitex Tunneling Packages Settings Help
Quick connect...
home/ubuntu/
- Name
- avo-devops-assignment
Remote monitoring
Follow terminal folder
UNREGISTERED VERSION - Please support MobaXterm by subscribing to the professional edition here: https://mobaxterm.mobatek.net

ubuntu@ip-172-31-62-202:~$ sudo apt install docker.io -y
Installing:
docker.io

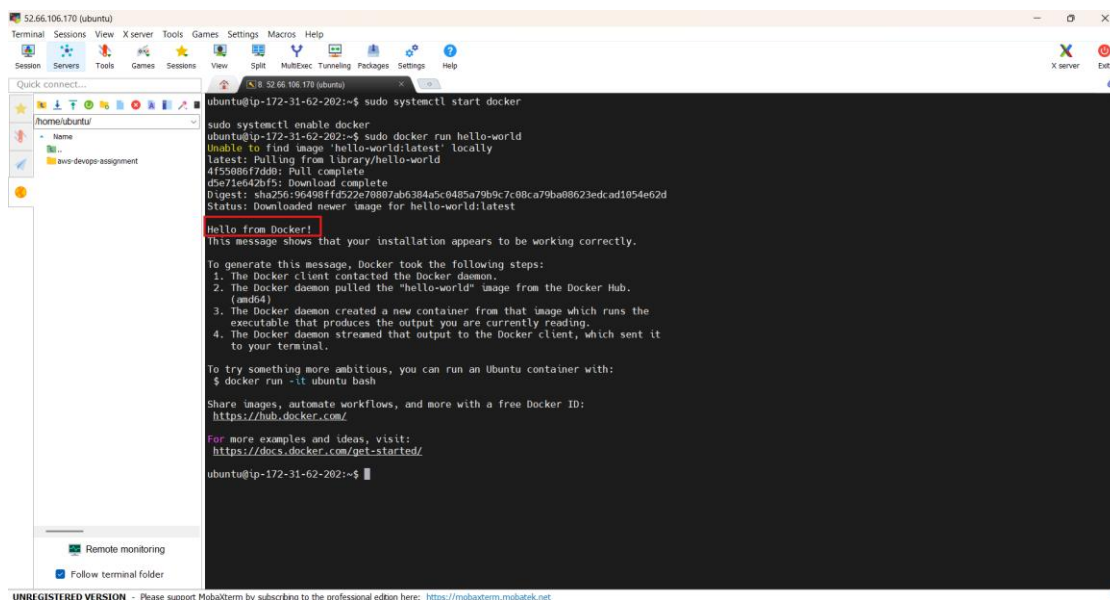
Installing dependencies:
bridge-utils containerd dns-root-data dnsmasq-base pigz runc ubuntu-fan

Suggested packages:
ifupdown cgroups-mount debotstrap docker-compose-v2 rinse zfs-fuse
aufs-tools | cgroup-lite docker-buildx docker-doc rootlesskit | zfsutils

Summary:
Upgrading: 0, Installing: 8, Removing: 0, Not Upgrading: 58
Download size: 74.0 MB
Space needed: 281 MB / 4677 MB available

Get:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute/universe amd64 pigz amd64 2.8-1build1 [68.5 kB]
Get:2 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute/main amd64 bridge-utils amd64 1.7.1-4ubuntu2 [34.8 kB]
Get:3 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute/main amd64 runc amd64 1.4.8-0ubuntu1 [9829 kB]
Get:4 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64 containerd amd64 2.2.2-0ubuntu1.1 [28.1 MB]
Get:5 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute/main amd64 dns-root-data all 2025080400build1 [6822 B]
Get:6 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64 dnsmasq-base amd64 2.92-1ubuntu0.3 [441 kB]
Get:7 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/universe amd64 docker.io amd64 29.1.3-0ubuntu4.1 [35.5 MB]
Get:8 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute/universe amd64 ubuntu-fan all 0.12.17 [34.3 kB]
Fetched 74.6 MB in 1s (71.2 MB/s)
Preconfiguring packages ...
Selecting previously unselected package pigz.
(Reading database ... 85351 files and directories currently installed.)
Preparing to unpack .../0-pigz_2.8-1build1_amd64.deb ...
Unpacking pigz (2.8-1build1) ...
Selecting previously unselected package bridge-utils.
Preparing to unpack .../1-bridge-utils_1.7.1-4ubuntu2_amd64.deb ...
Unpacking bridge-utils (1.7.1-4ubuntu2) ...
Selecting previously unselected package runc.
Preparing to unpack .../2-runc_1.4.8-0ubuntu1_amd64.deb ...
Unpacking runc (1.4.8-0ubuntu1) ...
Selecting previously unselected package containerd.
Preparing to unpack .../3-containerd_2.2.2-0ubuntu1.1_amd64.deb ...
Unpacking containerd (2.2.2-0ubuntu1.1) ...
Selecting previously unselected package dns-root-data.
Preparing to unpack .../4-dns-root-data_2025080400build1_all.deb ...
Unpacking dns-root-data (2025080400build1) ...
```

Figure 22: Docker Hello-World Output



```
52.66.106.170 (ubuntu)
Terminal Sessions View X server Tools Games Settings Macros Help
Session Servers Tools Games Sessions View Split Multitex Tunneling Packages Settings Help
Quick connect...
home/ubuntu/
- Name
- avo-devops-assignment
Remote monitoring
Follow terminal folder
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ubuntu@ip-172-31-62-202:~$ sudo systemctl start docker
sudo systemctl enable docker
ubuntu@ip-172-31-62-202:~$ sudo docker run hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
4f558867d98: Pull complete
25e71e4249f5: Download complete
Digest: sha256:96408ff522e78887ab63845c9485a79b9c7c08ca79ba88622edcad1054e62d
Status: Downloaded newer image for hello-world:latest

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
1. The Docker client contacted the Docker daemon.
2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
   (amd64)
3. The Docker daemon created a new container from that image which runs the
   executable that produces the output you are currently reading.
4. The Docker daemon streamed that output to the Docker client, which sent it
   to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
$ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
https://hub.docker.com/

For more examples and ideas, visit:
https://docs.docker.com/get-started/
ubuntu@ip-172-31-62-202:~$
```

Figure 23: Final Verification Screenshot

11. Conclusion

This assignment provided practical experience in deploying a web application on AWS using EC2 and Nginx. It improved my understanding of cloud infrastructure, Linux system administration, web server configuration, Git version control, and basic DevOps practices. The website was successfully deployed, tested, and documented as required.

12. Screenshots Included

The following screenshots are attached with the submission:

- AWS EC2 Dashboard
- EC2 Instance Running
- SSH Login Terminal
- Nginx Installation
- Nginx Status
- Linux Commands Output
- Website Running in Browser
- GitHub Repository
- Docker Hello World (Bonus)

Submission Details

GitHub Repository: <https://github.com/adhikarilaxman/aws-devops-assignment>

Submitted By: Laxman Adhikari